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INTERNATIONAL QUALIFICATIONS

# INTERNATIONAL A-LEVEL

# FURTHER MATHEMATICS

# FM05

(9665/FM05) Unit FM2 Mechanics

Mark scheme

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241XFM05/MS

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**Key to mark scheme abbreviations**

| | |
|---|---|
| **M** | Mark is for method |
| **m** | Mark is dependent on one or more M marks and is for method |
| **A** | Mark is dependent on M or m marks and is for accuracy |
| **B** | Mark is independent of M or m marks and is for method and accuracy |
| **E** | Mark is for explanation |
| **√ or ft** | Follow through from previous incorrect result |
| **CAO** | Correct answer only |
| **CSO** | Correct solution only |
| **AWFW** | Anything which falls within |
| **AWRT** | Anything which rounds to |
| **ACF** | Any correct form |
| **AG** | Answer given |
| **SC** | Special case |
| **oe** | Or equivalent |
| **A2, 1** | 2 or 1 (or 0) accuracy marks |
| **–*x* EE** | Deduct *x* marks for each error |
| **NMS** | No method shown |
| **PI** | Possibly implied |
| **SCA** | Substantially correct approach |
| **sf** | Significant figure(s) |
| **dp** | Decimal place(s) |
| **ISW** | Ignore subsequent working |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 1(a) | $2\begin{bmatrix}4\\7\end{bmatrix} + 6\begin{bmatrix}6\\-4\end{bmatrix} = 2\begin{bmatrix}7\\1\end{bmatrix} + 6\mathbf{v}_B$ | M1 | Applies conservation of momentum in a vector form. |
| | | A1 | Correct vector equation |
| | $6\mathbf{v}_B = \begin{bmatrix}30\\-12\end{bmatrix}$ | m1 | Solves their vector equation for velocity of $B$ |
| | $\mathbf{v}_B = \begin{bmatrix}5\\-2\end{bmatrix}\ [\text{m s}^{-1}]$ | A1 | Correct velocity |
| | | 4 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 1(b) | $\mathbf{I} = 2\begin{bmatrix}7\\1\end{bmatrix} - 2\begin{bmatrix}4\\7\end{bmatrix}$ | M1 | Uses impulse equation with vectors |
| | $= \begin{bmatrix}6\\-12\end{bmatrix}$ | A1 | Correct impulse |
| | $I = \sqrt{6^2 + 12^2}$ | M1 | Finds magnitude of their impulse |
| | $= 6\sqrt{5}\ [\text{N s}]$ | A1 | Correct magnitude of the impulse |
| | | 4 | |

| | | | |
|---|---|---|---|
| | **Question 1 Total** | **8** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 2(a) | $4 = \frac{2\pi}{\omega}$<br>$\omega = \frac{\pi}{2}$ | M1 | Correct value of $\omega$ |
| | $v_{\max} = \frac{3}{4} \times \frac{\pi}{2} = \frac{3\pi}{8} \quad \left[\text{m s}^{-1}\right]$ | A1 | Correct speed **oe** in terms of $\pi$<br>(eg $0.375\pi$) |
| | | 2 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 2(b) | $v^2 = \omega^2\left(a^2 - x^2\right)$<br>$v^2 = \left(\frac{\pi}{2}\right)^2\left(0.75^2 - 0.45^2\right)$ | M1 | Uses SHM speed formula or other SHM method with their amplitude and consistent displacement. |
| | $v = \frac{3\pi}{10} \quad \left[\text{m s}^{-1}\right]$ | A1ft | Correct speed for their amplitude.<br>**oe** in terms of $\pi$<br>(eg $0.3\pi$) |
| | | 2 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 2(c) | $x = \frac{3}{4}\cos\left(\frac{\pi t}{2}\right)$ | M1 | Finds expression for displacement in terms of time |
| | $0.45 = \frac{3}{4}\cos\left(\frac{\pi t}{2}\right)$ | A1ft | Correct equation for time, for their amplitude. |
| | $t = \frac{2}{\pi}\cos^{-1}\left(\frac{3}{5}\right)$ | | |
| | $t = 0.5903 \ [\text{seconds}]$ | A1 | Correct time |
| | | 3 | |

| | | | |
|---|---|---|---|
| | **Question 2 Total** | **7** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 3(a) | $8\cos 60° = v\cos 30°$ | M1 A1 | **M1**: Equation for motion parallel to wall with at least one side correct<br>**A1**: Fully correct |
| | $4 = \frac{\sqrt{3}}{2}v$ | | |
| | $v = \frac{8}{\sqrt{3}} = \frac{8\sqrt{3}}{3}$ | A1 | Correct value for $v$ **ACF** |
| | | 3 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 3(b) | $e \times 8\sin 60° = v\sin 30°$ | M1 A1ft | **M1**: Equation for motion perpendicular to wall with at least one side correct<br>**A1ft**: Fully correct<br>**ft** their $v$ |
| | $e \times 4\sqrt{3} = \frac{4\sqrt{3}}{3}$ | | |
| | $e = \frac{1}{3}$ | A1 | Correct value for $e$ **ACF** |
| | | 3 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 3(c) | $I = 0.2 \times \frac{4\sqrt{3}}{3} - 0.2 \times \left(-4\sqrt{3}\right)$ | M1 A1 | **M1**: Impulse equation with correct values and any signs<br>**A1**: Fully correct |
| | $= \frac{16\sqrt{3}}{15}$ [N s] | A1 | Correct impulse **ACF** |
| | | 3 | |

| | | | |
|---|---|---|---|
| | **Question 3 Total** | **9** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| **4** | At highest point the tension in the string is zero when $U$ is the minimum possible value | | |
| | $m \times 9.8 = \frac{mv^2}{0.8}$ | **M1** | Equation for motion at highest point |
| | $v^2 = 7.84$ | | |
| | $v = 2.8$ | **A1** | Correct speed at highest point ($v^2 = rg$) |
| | Conservation of energy $\frac{1}{2} \times mU^2 = \frac{1}{2} \times m \times 7.84 + m \times 9.8 \times 1.6$ | **M1 A1** | **M1**: Three term equation for conservation of energy<br>**A1**: Fully correct |
| | $U^2 = 39.2$ | | ($U^2 = 5rg$) |
| | $U = \frac{14\sqrt{5}}{5} = 6.3$ [to 2sf] | **A1** | Correct value for $U$ **AWRT** 6.3 |

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|---|---|---|---|
| | **Question 4 Total** | **5** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 5 | $m\frac{\mathrm{d}v}{\mathrm{d}t} = mg\sin 30° - \mu \times mg\cos 30° - mkv$<br>$\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{g}{2} - \mu g\frac{\sqrt{3}}{2} - kv$ | **M1** | Four term differential equation from Newton's second law |
| | $\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{1}{2}\left(g - \mu g\sqrt{3} - 2kv\right)$ | **A1** | Correct differential equation |
| | $\int \frac{1}{\left(2kv - g + \mu g\sqrt{3}\right)}\,\mathrm{d}v = \int -\frac{1}{2}\,\mathrm{d}t$ | **M1** | Uses separation of variables or integrating factor |
| | $\frac{1}{2k}\ln\left(2kv - g + \mu g\sqrt{3}\right) = -\frac{1}{2}t + c$ | **m1**<br>**A1** | Integrates<br>Correct integration |
| | $v = 0, t = 0 \Rightarrow c = \frac{1}{2k}\ln\left(\mu g\sqrt{3} - g\right)$ | **A1** | Finds correct constant of integration |
| | $\ln\left(2kv - g + \mu g\sqrt{3}\right) = -kt + \ln\left(\mu g\sqrt{3} - g\right)$<br>$\frac{2kv - g + \mu g\sqrt{3}}{\mu g\sqrt{3} - g} = \mathrm{e}^{-kt}$ | **M1** | Solves a logarithmic equation for $v$ |
| | $v = \frac{g}{2k}\left(1 - \mu\sqrt{3}\right)\left(1 - \mathrm{e}^{-kt}\right)$ | **A1** | Correct expression for $v$<br>**ACF** not involving trigonometric functions |

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|---|---|---|---|
| | **Question 5 Total** | **8** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 6(a) | Let $\theta$ be the angle between the string and the vertical at time $t$ | | |
| | $m\times 1.4\,\ddot{\theta} = -m\times 9.8\sin\theta$ | M1 | Forms differential equation |
| | | A1 | Correct differential equation |
| | $\sin\theta \approx \theta$ | B1 | Uses small angle approximation |
| | $\ddot{\theta} \approx -\frac{9.8}{1.4}\theta = -7\theta$ | A1 | Simplifies to SHM form |
| | $\therefore$ SHM | | Accept $\ddot{\theta} \approx -\frac{g}{l}\theta$ |
| | As acceleration proportional to the displacement and in the opposite sense | E1 | Concludes that motion is SHM from correct working |
| | | 5 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 6(b) | Period $= \frac{2\pi}{\sqrt{7}} = \frac{2\sqrt{7}}{7}\pi \quad [= 2.3748...]$ | B1 | Correct period stated during working |
| | Total angle for 1.8 metres<br>$= \frac{1.8}{1.4} = \frac{9}{7} \quad [= 1.286 \text{ radians}]$ | M1 | Finds total angle for the motion |
| | Angle per complete period<br>$= 4 \times \frac{\pi}{20} = \frac{\pi}{5} \quad [= 0.6283... \text{ radians}]$<br>$\left(\frac{9}{7}\right) \div \left(\frac{\pi}{5}\right) = \frac{45}{7\pi} \quad [= 2.04627...]$ | A1 | Obtains exact value or **AWRT** 2.047 |
| | [So between 2 and 3 periods required.]<br>$\frac{9}{7} - \frac{2\pi}{5} \quad [= 0.029077... \text{ radians}]$ | A1 | Correct angle for remaining motion since last complete period or correct distance (0.04071…) |
| | $\theta = \frac{\pi}{20} - \left(\frac{9}{7} - \frac{2\pi}{5}\right) = \frac{9\pi}{20} - \frac{9}{7} \quad [= 0.12800...]$ | M1<br>A1 | Finds $\theta$<br>Correct $\theta$ |
| | $\frac{9\pi}{20} - \frac{9}{7} = \frac{\pi}{20}\cos\left(\sqrt{7}t\right)$<br>$t = 0.23368...$ seconds | M1 | Finds time for motion since last period |
| | Total Time $= 2 \times \frac{2\pi}{\sqrt{7}} + 0.23368$<br>$= 4.98332...$<br>$= 5.0$ seconds | A1 | Correct total time<br>**AWRT** 5.0 |
| | | 8 | |

| | | | |
|---|---|---|---|
| | **Question 6 Total** | 13 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 7(a) | $e = \sqrt{5^2 + 3^2} - 5$<br>$= \sqrt{34} - 5$ | M1<br>A1 | Finds extension of strings<br>Correct extension |
| | $\text{EPE} = \dfrac{800 \times \left(\sqrt{34} - 5\right)^2}{2 \times 5}$<br>$= 55.238...$ | M1 | Uses EPE formula with their extension |
| | Total EPE $= 110$ [J] | A1 | Correct total EPE |
| | | 4 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 7(b) | Strings slack when height above C is 1 metre | B1 | Finds height when strings at natural length<br>**PI** |
| | Change in GPE $= 7 \times 9.8 \times 1$<br>$= 68.6$ | B1 | Correct change in GPE |
| | $110.47... - 68.6 = \dfrac{1}{2} \times 7 \times v^2$ | M1 A1 | **M1**: Three term energy equation<br>**A1**: Fully correct |
| | $v^2 = 11.971$<br>$v = 3.4599...$<br>$= 3.5$ [m s$^{-1}$ to 2 sf] | A1 | **AWRT** 3.5<br>Allow **AWRT** 3.4 if 110 J used |
| | | 5 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 7(c) | $110.5 = 7 \times 9.8h$ | M1 | Two term energy equation |
| | $h = 1.6$ [m] (to 2 sf) | A1 | Correct height |
| | | 2 | |

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|---|---|---|---|
| | **Question 7 Total** | **11** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 8(a) | $0 = 20\sin 35°t - \frac{1}{2}g\cos 30°t^2$ | M1 | Equation for motion perpendicular to the plane |
| | $t = \frac{40\sin 35°}{g\cos 30°}$ $[= 2.703...]$ | A1 | Correct equation |
| | | A1 | Correct time |
| | $\dot{x} = 20\cos 35° - g\sin 30° \times \frac{40\sin 35°}{g\cos 30°}$ | M1 | Finds both components of velocity |
| | $= 3.1368...$ | A1 | Correct parallel component |
| | $\dot{y} = 20\sin 35° - g\cos 30° \times \frac{40\sin 35°}{g\cos 30°}$ | | |
| | $= -20\sin 35°$ | | |
| | $= -11.4715...$ | A1 | Correct perpendicular component |
| | $v = \sqrt{11.4715...^2 + 3.1368...^2}$ | | |
| | $= 11.89...$ | | |
| | $= 12$ $\left[\text{m s}^{-1}\right]$ (to 2 sf) | A1 | Correct speed |
| | | 7 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 8(b) | Require rebound angle to be greater than 60° | B1 | Identifies 60° |
| | $\tan\theta = \frac{11.4715... \times e}{3.1368...}$ | M1 | Forms expression for tan of the rebound angle |
| | $\frac{11.4715... \times e}{3.1368...} > \sqrt{3}$ | A1 | Correct inequality |
| | $0.47 < e < 1$ | A1 | Correct range<br>Allow lower limit between 0.46 and 0.48<br>Accept $0.47 \le e \le 1$ |
| | | 4 | |

| | | | |
|---|---|---|---|
| | **Question 8 Total** | **11** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 9(a) | $12\times 8\cos 60^\circ = 12v_A + 3v_B$<br>$16 = 4v_A + 1v_B$ | M1 | Equation for conservation of momentum |
| | $v_B - v_A = -e(0 - 8\cos 60^\circ)$ | M1 | Equation for coefficient of restitution |
| | $v_B - v_A = 4e$ | A1 | Two correct equations |
| | $v_A = \dfrac{16-4e}{5}$ | A1 | Correct component of *A* along line of centres |
| | $V^2 = (8\sin 60^\circ)^2 + \left(\dfrac{16-4e}{5}\right)^2$ | M1 | Finds expression for speed of *A* after the collision |
| | $V = \sqrt{48 + \left(\dfrac{16-4e}{5}\right)^2}$ | A1 | Correct expression **ACF** |
| | | 6 | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 9(b) | $V_{\min} = \sqrt{(8\sin 60^\circ)^2 + \left(\dfrac{16-4}{5}\right)^2} = \dfrac{8\sqrt{21}}{5}$ | M1 | Uses $e = 1$ |
| | Maximum change in speed = $8 - \dfrac{8\sqrt{21}}{5} = \dfrac{40 - 8\sqrt{21}}{5}$ | A1 | Correct maximum change in speed **ACF** |
| | | 2 | |

| | | | |
|---|---|---|---|
| | **Question 9 Total** | 8 | |